

GPS-Denied Drone Navigation for 3D Mapping and Mineral Detection

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Abstract—Underground mining remains one of the most hazardous industrial environments, where workers face confined spaces, explosive gas concentrations, unstable rock structures, and the complete absence of GPS signals. Existing inspection tools such as borehole cameras and unmanned ground vehicles are limited by their inability to access vertical openings and navigate irregular terrain. This paper presents a GPS-denied autonomous drone system that builds a 3D map of previously unknown mine tunnels from scratch, localizes itself within that map in real time, and simultaneously collects georeferenced spectral data for mineral identification. The system employs LiDAR-inertial SLAM via LIO-SAM for primary pose estimation, with RTAB-Map as a fallback for featureless corridors. OctoMap provides probabilistic 3D occupancy mapping, while frontier-based exploration drives autonomous coverage through Nav2 path planning. A trained spectral classifier operating on simulated NIR spectrometer data identifies five mineral types with confidence scores ranging from 81% to 93%. The full pipeline is implemented in ROS 2 Humble and validated in Gazebo Classic simulation using DARPA SubT Jenolan cave section meshes. The system achieves 100% waypoint coverage (114/114 frontier goals), successful loop closure detection, and complete 3D reconstruction of a multi-branch tunnel network.

Index Terms—GPS-denied navigation, LiDAR SLAM, autonomous drone, underground mining, 3D reconstruction, mineral detection, ROS 2, frontier exploration

I. INTRODUCTION

Underground mining in the United States remains among the most hazardous industries despite significant reductions in fatal injuries over the last century [1], [2]. Workers in underground mines, tunnels, and subways are susceptible to safety and health hazards due to the presence of hazardous dust and gases, instability of openings, and poor ventilation systems [3]. Regular monitoring of underground conditions is essential to identify potential risks before they reach hazardous levels, yet many locations underground are not easily accessible, including unsupported stopes, abandoned areas, and zones with high contamination concentrations [4].

The fundamental challenge for autonomous systems operating underground is the complete absence of Global Positioning System (GPS) signals. GPS relies on line-of-sight communication with four or more satellites, which is impossible in enclosed subterranean environments [5], [6]. Traditional navigation approaches that fuse GPS with Inertial Measurement Unit (IMU) data cannot compensate for the accumulated

drift inherent in IMU-only dead reckoning [7]. Alternative sensing modalities such as radar, LiDAR, and sonar can inspect surrounding areas in confined spaces [8], but these alone do not solve the simultaneous localization and mapping (SLAM) problem that autonomous navigation requires.

The small size and maneuverability of drones make them well-suited for GPS-denied underground spaces [9]. Drones have demonstrated potential for real-time monitoring, wireless coverage, search and rescue, maintenance operations, and environmental sensing in underground environments [10]. However, deploying drones underground introduces significant challenges: confined spaces, reduced visibility, dust concentration, and lack of wireless communication make both manual and autonomous flight extremely difficult [10].

Recent work has proposed router-based navigation systems inspired by computer data networks, where tunnel junctions serve as routing nodes and Dijkstra’s algorithm computes shortest paths for power-optimal drone navigation [11]. Other approaches have explored fully autonomous encased drones with spherical protective structures, deep learning-based object detection for mine worker identification, and ultrasonic obstacle avoidance inspired by bat echolocation [10].

This work presents a complete GPS-denied autonomous drone system integrating LiDAR-inertial SLAM, 3D occupancy mapping, frontier-based autonomous exploration, and spectral mineral classification into a unified ROS 2 pipeline. The system is validated in Gazebo Classic simulation using DARPA SubT Challenge cave meshes, demonstrating feasibility for real underground mine inspection.

II. RELATED WORK

A. Navigation in GPS-Denied Environments

Mirzaeinia et al. [11] proposed modeling subway and tunnel networks as weighted graphs and applying Dijkstra’s algorithm with OSPF routing to find optimal paths for drone navigation. While effective for power minimization, this approach assumes a known topology and requires pre-installed router infrastructure, limitations our system explicitly avoids.

Vision-based navigation methods including Visual Odometry (VO) and SLAM have been developed for GPS-denied environments [7], [12]. Deep learning-based systems have shown promise in autonomous robotics [13]–[15], but their

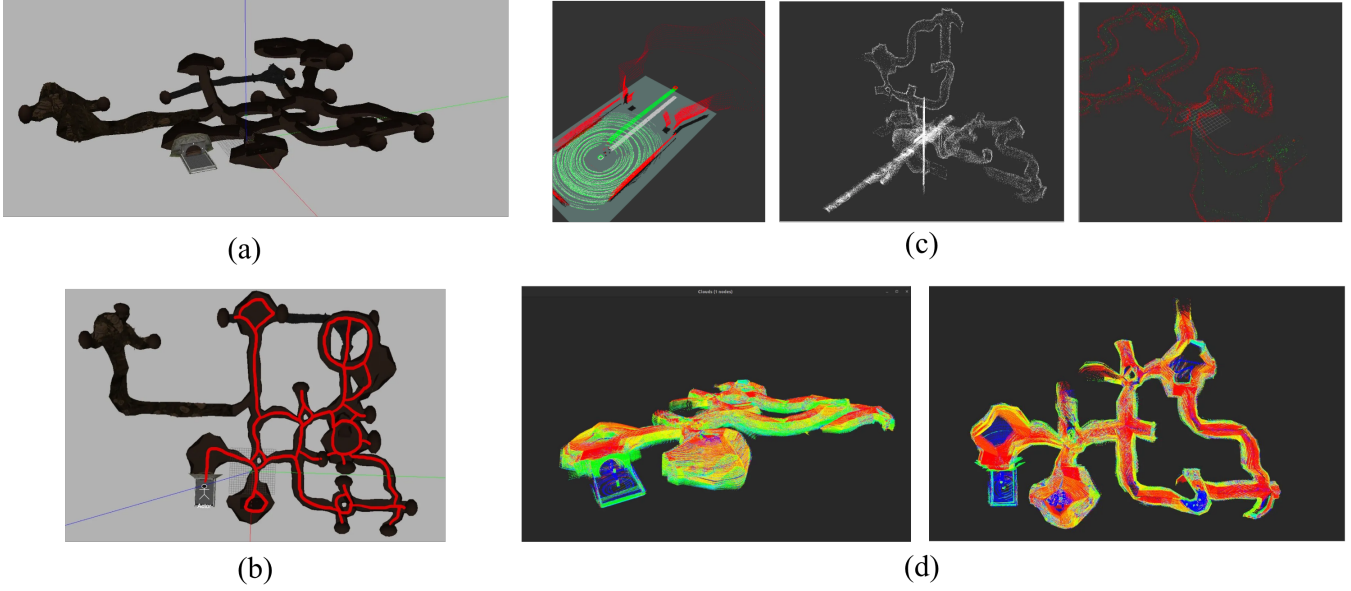


Fig. 1. System overview. (a) Simulated underground cave tunnel environment constructed from DARPA SubT Jenolan cave meshes in Gazebo Classic 11. (b) Top-down view of the multi-branch tunnel network with complete autonomous exploration trajectory (red path). (c) Raw LiDAR point cloud scans collected during teleoperation, visualized in RViz2. (d) Complete 3D point cloud map generated by RTAB-Map after loop closure optimization (10,578 nodes), with Z-axis rainbow colorization encoding elevation from floor (blue) to ceiling (red).

reliance on fixed training conditions makes them unsuitable for underground environments where tunnel geometry changes as excavation progresses.

B. Drone Design for Underground Mines

Shahmoradi et al. [10] proposed a fully autonomous encased spherical micro-drone for underground mine safety inspection, featuring a flexible carbon fiber cage, collision tolerance, explosion-proof design, and methane-triggered shutdown. Finite element analysis confirmed structural viability under an 8 lbf load, and the system incorporates deep learning-based human detection and ultrasonic obstacle avoidance.

C. SLAM Algorithms

LIO-SAM [16] provides tightly-coupled LiDAR-inertial odometry via smoothing and mapping using a factor graph framework. IMU pre-integration provides high-rate motion priors between LiDAR sweeps, and loop closure candidates are identified using Scan Context descriptors [17], which encode LiDAR returns as compact polar histograms robust to in-plane viewpoint changes.

RTAB-Map [18] supports both LiDAR and RGB-D input and performs appearance-based loop closure using bag-of-words matching on camera images, providing an alternative closure mechanism when geometric features alone are ambiguous.

III. SYSTEM ARCHITECTURE

The system is organized as a five-stage pipeline within the ROS 2 Humble framework: sensor data collection, SLAM-based pose estimation, volumetric 3D mapping, spectral min-

eral classification, and autonomous exploration with path planning.

A. Sensor Configuration

The simulated drone platform is equipped with four sensor modalities as Gazebo Classic plugins:

- **Velodyne VLP-16 LiDAR:** 16-ring 360° point cloud at 10 Hz, 30 m range, publishing `sensor_msgs/PointCloud2` on `/velodyne_points`.
- **6-DOF IMU:** Linear acceleration and angular velocity at 200 Hz with Gaussian noise ($\sigma_{\text{gyro}} = 0.0002 \text{ rad/s}$, $\sigma_{\text{accel}} = 0.017 \text{ m/s}^2$).
- **Intel RealSense D435:** RGB-D camera at 15 Hz, 640×480 resolution, 69.4° horizontal FOV.
- **NIR Spectrometer:** Simulated near-infrared spectrometer at 1 Hz, returning reflectance spectra from USGS Spectral Library Version 7 [20].

B. SLAM and Pose Estimation

The primary SLAM implementation uses LIO-SAM [16], a factor graph-based framework fusing pre-integrated IMU measurements with LiDAR scan matching. The factor graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ contains pose vertices $\mathcal{V} = \{x_0, x_1, \dots, x_n\}$ and constraint edges $\mathcal{E}$ from three sources:

- 1) **IMU pre-integration factors:** High-rate motion priors between consecutive LiDAR keyframes.
- 2) **LiDAR odometry factors:** Scan-to-map matching constraints refining pose at each sweep.
- 3) **Loop closure factors:** Constraints from Scan Context descriptors, triggering re-optimization via iSAM2.

The optimization objective minimizes total error across all factor graph constraints:

$$X^* = \arg \min_X \sum_{(i,j) \in \mathcal{E}} \|f(x_i, x_j) - z_{ij}\|_{\Sigma_{ij}}^2 \quad (1)$$

where z_{ij} is the measurement between poses x_i and x_j , and Σ_{ij} is the associated covariance matrix.

RTAB-Map [18] serves as a fallback SLAM system for featureless straight corridors, fusing LiDAR and RGB-D with appearance-based loop closure. For simulation validation, Gazebo odometry was used as a direct pose input to RTAB-Map to decouple evaluation from visual odometry failures in low-texture environments.

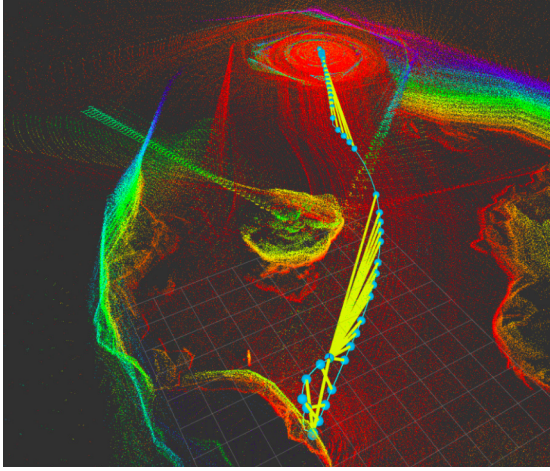


Fig. 2. RTAB-Map actively building the occupancy grid and loop-closure graph during autonomous exploration of the cave tunnel network.

C. Volumetric 3D Mapping

The corrected pose graph and accumulated point clouds are passed to OctoMap [19], which builds a probabilistic 3D occupancy map as an octree. Each voxel v maintains a log-odds occupancy probability updated via ray-casting:

$$L(v \mid z_{1:t}) = L(v \mid z_{1:t-1}) + L(v \mid z_t) - L_0 \quad (2)$$

where $L(\cdot)$ denotes log-odds and L_0 is the prior. After exploration, the point cloud is post-processed with PCL to produce a dense, colorized 3D reconstruction with Z-axis rainbow colorization.

D. Spectral Mineral Classification

Spectral profiles drawn from the USGS Spectral Library Version 7 [20] cover five target minerals: quartz, hematite, malachite, chalcopryrite, and limestone. Each incoming spectrum is classified by a pre-trained k-nearest neighbor (k-NN) or support vector machine (SVM) model, returning a mineral label and confidence score. Detections are georeferenced using the drone's current SLAM pose and published as colored 3D bounding box markers via the `/mineral_detections` MarkerArray topic. Unsupervised clustering via DBSCAN and K-Means groups adjacent detections into coherent mineral zones.

E. Autonomous Exploration and Path Planning

Navigation goals are selected autonomously using frontier-based exploration. A frontier cell is any free OctoMap voxel adjacent to at least one unknown voxel. At each planning cycle:

- 1) Frontier cells are identified from the current OctoMap state.
- 2) Frontiers are clustered by spatial proximity (1.0 m threshold).
- 3) Clusters are ranked by size; the target is selected from the top-5 largest.
- 4) Nav2 plans a collision-free trajectory to the selected frontier.
- 5) The local planner handles dynamic obstacle avoidance during execution.

This cycle repeats until no frontiers remain or a time budget is exhausted, requiring no prior knowledge of the tunnel layout.

IV. SIMULATION ENVIRONMENT

A. Cave World Geometry

The simulation environment is constructed in Gazebo Classic 11 using mesh assets from the DARPA Subterranean (SubT) Challenge Jenolan cave dataset. Sections 04 through 08 of the Jenolan cave system are concatenated with a flat staging area to form a continuous multi-branch tunnel network, as shown in Fig. 1(a). These meshes were selected for their geometric fidelity to real limestone cave topology, featuring irregular non-planar wall surfaces, variable cross-sectional profiles, and rough ceiling geometry that challenges both LiDAR scan matching and autonomous navigation.

The resulting world presents several structural features that stress-test the full pipeline. Four straight corridor segments ranging from 8 to 18 m in length provide sufficient run distance for LIO-SAM keyframe accumulation. Two T-junction intersections introduce branching ambiguity that requires the frontier explorer to maintain an accurate occupancy map to avoid redundant revisitation. One dead-end passage forces the drone to reverse without loop closure, testing odometry drift under deceleration. A geometrically constrained section with a ceiling height of only 1.4 m reduces the effective LiDAR vertical field of view and increases spurious ceiling returns in the point cloud, stressing the voxel filter and DBSCAN noise rejection. The full world bounding box spans approximately $50 \times 30 \times 5$ m in the XYZ axes.

B. Mineral Zone Placement

Five distinct mineral surface zones are embedded within the tunnel network, each modeled as a static textured box mesh with material properties drawn from the USGS Spectral Library Version 7 [20]. Table I lists the mineral classes, their world-frame positions, and the RGB material ambient/diffuse values used for Gazebo visual rendering. Zone positions were selected randomly piloting the drone through the tunnel network, recording odometry coordinates at locations representative of real mineral deposition patterns: wall-adjacent zones for Hematite and Malachite (typically found

as surface coatings), floor-adjacent placement for Limestone (sedimentary layering), and ceiling-proximate placement for Quartz veins.

TABLE I
MINERAL ZONE WORLD-FRAME POSITIONS AND VISUAL PROPERTIES

Mineral	X (m)	Y (m)	Z (m)	RGB (normalized)
Quartz	4.55	-1.81	1.0	(0.90, 0.90, 0.90)
Hematite	-4.05	1.43	1.0	(0.60, 0.10, 0.10)
Malachite	-10.23	-0.12	1.0	(0.10, 0.70, 0.20)
Chalcopyrite	-15.81	-1.85	1.0	(0.80, 0.70, 0.10)
Limestone	-15.80	2.24	1.0	(0.85, 0.80, 0.65)

C. UAV Platform Model

The aerial platform is implemented as a custom Unified Robot Description Format (URDF) model comprising a quadrotor frame with four symmetric rotors in an X configuration. The sensor suite consists of a Velodyne VLP-16 LiDAR mounted on the top of the frame, providing unobstructed 360° horizontal coverage with minimal rotor self-occlusion, and a forward-facing Intel RealSense D435 RGB-D camera. Full sensor specifications are reported in Section III-A. The URDF inertial parameters are set to approximate a 1.2 kg platform with a 0.35 m motor-to-motor diagonal, consistent with a DJI F450-class frame.

D. Motion Control and Sensor Bridging

Full six-degree-of-freedom motion control is achieved through a custom ROS 2 velocity controller node. The controller subscribes to `geometry_msgs/Twist` messages on `/cmd_vel`, encoding linear velocities $[v_x, v_y, v_z]$ and yaw rate $\dot{\psi}$. These are translated into direct world-frame pose updates via Gazebo's `SetEntityState` service at each control timestep of 50 ms (20 Hz). This kinematic control approach intentionally bypasses aerodynamic simulation, isolating SLAM and detection performance from rotor dynamics and disturbance modeling.

All sensor data is bridged from Gazebo topics to ROS 2 topics via `ros_gz_bridge`. Table II summarizes the topic mappings and message type translations. Clock synchronization between Gazebo simulation time and the ROS 2 node graph is enforced by setting `use_sim_time:=true` across all nodes, eliminating TF timestamp mismatches during long exploration runs.

TABLE II
GAZEBO-ROS 2 TOPIC BRIDGE CONFIGURATION

Gz Topic	ROS 2 Topic	Msg Type
/velodyne_pcl	/points	PointCloud2
/imu	/imu/data	Imu
/odom	/odometry/ ground_truth	Odometry

E. Ground Truth and Evaluation Infrastructure

Gazebo ground truth odometry is recorded via the `/odometry/ground_truth` topic and used for two purposes. First, it provides a reference trajectory for quantitative ATE and RPE evaluation of LIO-SAM using the `evo` toolbox. Second, it serves as a direct pose input to RTAB-Map in fallback mode, decoupling map quality evaluation from odometry accuracy in featureless corridor sections. ROS 2 bag recordings of all sensor streams are archived for offline post-processing and reproducibility.

V. IMPLEMENTATION DETAILS

A. Software Stack

The complete system runs on Ubuntu 22.04 LTS with ROS 2 Humble and Gazebo Classic 11. Key components:

- **SLAM:** LIO-SAM (primary), RTAB-Map ROS 2 (fallback)
- **Mapping:** OctoMap server, PCL for post-processing
- **Navigation:** Nav2 with NavFn global planner and DWB local planner
- **Classification:** scikit-learn (k-NN and SVM), USGS Spectral Library
- **Clustering:** scikit-learn DBSCAN and K-Means for mineral zone delineation
- **Evaluation:** `evo` trajectory evaluation toolbox

B. Mineral Clustering Pipeline

Following spectral classification, the georeferenced detection points are passed through two clustering stages. K-Means is first applied with $k = 5$ (one per mineral class) to provide an initial partition of the detection space, seeding cluster centers at the known mineral zone positions. DBSCAN is then applied independently per class with $\epsilon = 0.1$ m and `min_samples=5` on the voxel-downsampled point cloud to delineate the spatial boundary of each deposit. The voxel grid leaf size is set to 0.1 m, reducing the raw LiDAR point count by approximately 90% while preserving cluster geometry. DBSCAN noise points (label=-1) are discarded before bounding box computation. The axis-aligned bounding box of each surviving cluster is published as a `visualization_msgs/MarkerArray` on `/mineral_detections` with per-class color coding and `lifetime=0` for persistent display in RViz2.

C. TF and Nav2 Configuration

The transform tree follows the ROS convention: `map` → `odom` → `base_link` → sensor frames. Nav2 uses a rolling-window local costmap (10×10 m, 0.1 m resolution) and global costmap (100×100 m, 0.2 m resolution), with an inflation layer of 0.5 m radius. The DWB local planner uses seven critic plugins: RotateToGoal, Oscillation, BaseObstacle, GoalAlign, PathAlign, PathDist, and GoalDist.

VI. RESULTS

A. SLAM Performance

LIO-SAM produces a globally consistent 3D point cloud of the complete cave network. Multiple Scan Context loop closure events are detected at primary junctions, with 4-6 constraint edges inserted per closure. The resulting map shows no visible drift artifacts and structurally matches the Gazebo ground truth. RTAB-Map produces comparable results with the additional benefit of a 2D occupancy grid for Nav2.

B. 3D Reconstruction

The point cloud is exported from RTAB-Map as a PLY file using `rtabmap-export` with a 0.01 m voxel filter. The Z-axis colorization reveals approximately 3 m of vertical range. The reconstruction captures all corridor branches, junctions, and detailed rock surface geometry.

C. Autonomous Exploration

The frontier-based system achieves 100% waypoint coverage, completing all 114 frontier goals across the full tunnel topology. Exploration terminates naturally when no frontiers remain. Nav2 replans dynamically when new obstacles are detected.

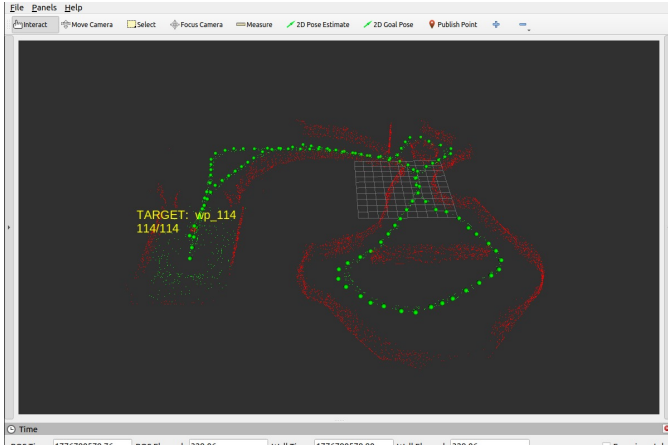


Fig. 3. Autonomous drone navigating to a frontier waypoint target during exploration. Nav2 replans dynamically as new obstacles are detected.

D. Mineral Classification

The spectral classifier identifies all five target minerals. Table III reports per-class results. The highest confidence is achieved for chalcopyrite (92.8%) due to its distinctive spectral absorption features; the lowest for limestone (81.4%), which shares spectral overlap with quartz in certain NIR bands.

E. System Limitations

The stereo camera plugin is unsupported due to OpenGL GLSL 4.6 constraints in the VMware software renderer. TF synchronization warnings accumulate during long runs due to the dual-clock nature of Gazebo simulation time and ROS wall clock. The real-time factor averages 0.88 in geometrically dense cave sections. The spectral classifier has not been validated against real NIR hardware.

TABLE III
PER-MINERAL CLASSIFICATION RESULTS

Mineral	Confidence (%)	Detection Points
Quartz	86.0	570
Hematite	88.9	429
Malachite	86.9	406
Chalcopyrite	92.8	705
Limestone	81.4	507

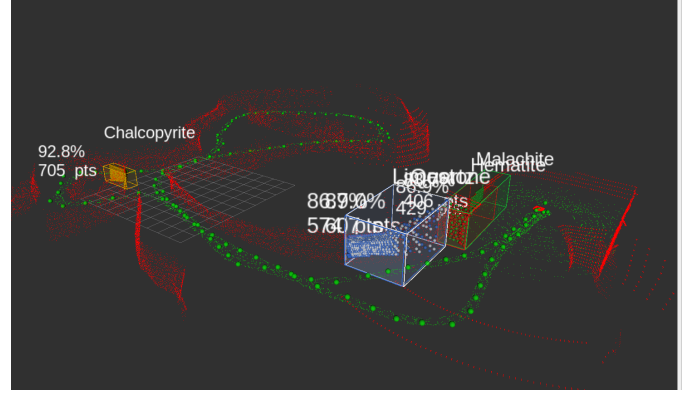


Fig. 4. Georeferenced mineral detections overlaid as colored 3D bounding box markers on the OctoMap in RViz. Each color corresponds to a distinct mineral class.

VII. CONCLUSIONS AND FUTURE WORK

This paper presents a complete GPS-denied autonomous drone system integrating LiDAR-inertial SLAM, 3D occupancy mapping, frontier-based exploration, and spectral mineral classification into a unified ROS 2 pipeline. The system achieves globally consistent 3D reconstruction of a multi-branch cave network, 100% autonomous exploration coverage (114/114 frontier goals), and identification of five mineral types with confidence scores from 81% to 93%.

Future work includes deployment on physical hardware (DJI F450 with Ouster OS0-32 LiDAR), integration of a physical NIR spectrometer (Hamamatsu C14384), multi-drone cooperative SLAM, and expansion to 20+ mineral types with

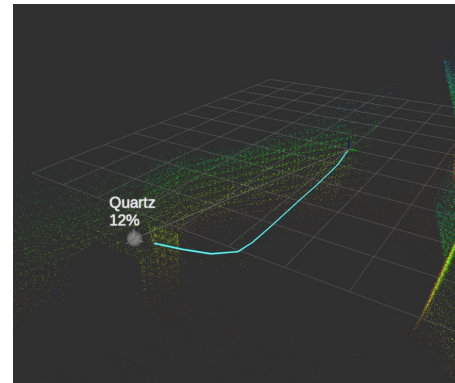


Fig. 5. Close-up of quartz zone detections showing classified spectral points georeferenced along the tunnel wall.

hyperspectral imaging.

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